# Abdelrahman Khalifa

Systems Software Engineer | Rust, C, Linux

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## SUMMARY

Systems software engineer working in Rust, C and modern C++ on Linux. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting): built the Printers page for COSMIC Settings and landed 7 PRs upstream in cups-rs and libcups. Comfortable with FFI, IPC, async code and upstream code review.

## EXPERIENCE

**Google Summer of Code Contributor** | **The Linux Foundation (OpenPrinting)** | **May 2026 – Oct 2026**

*Open-source project: Printers page for COSMIC Settings, in Rust; passed the final evaluation | github.com/Abd002/GSOC-2026 | github.com/Abd002/cosmic-printers*

- Built the **Printers page for COSMIC Settings** and a standalone app sharing the same UI: printers grouped by device, status and supply levels, print queue control, test pages, live updates, and an Add Printer flow that sets up legacy printers through Printer Applications (PAPPL) instead of PPD drivers (cosmic-settings #2182, in review).
- Designed the stack as 5 Rust crates: `printers-core` (types, grouping queues into physical devices), `printers-client` (async varlink client), `printers-server` (CUPS/IPP), `printers-ui` (iced) and `printers-app`.
- Ran the backend as a varlink IPC service in `cosmic-settings-daemon` (cosmic-settings-daemon #188, in review), so the C FFI code never loads in the Settings process; the standalone app falls back to running it in-process.
- Tested on real HP printers and Printer Applications (Ghostscript, Gutenprint, HPLIP); CI checks formatting, Clippy and tests on several architectures and runs CUPS, Avahi and `ippeveprinter`.

**Embedded Software Engineer** | **EKSON** | **Aug 2025 – Present**

*Sole software engineer for 3-DoF and 6-DoF pneumatic motion simulators*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS: PID valve control with IMU feedback, safety stops and a UART protocol.
- Built the PyQt5 desktop control app, cutting CPU usage and latency by about 40% (measured) using asyncio to wait for ESP32 and server messages concurrently; added remote control and firmware updates through a Node.js/MongoDB backend.

**Embedded Linux Systems Intern** | **Orange Digital Center (ODC)** | **Dec 2023 – Jan 2024**

- Built Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

## OPEN SOURCE PROJECTS

**Upstream contributions — OpenPrinting cups-rs (Rust) and libcups (C)** | **2026**

- cups-rs: ported to libcups3 with CUPS 2 and CUPS 3 builds and CI; added IPP management operations, Printer Application support, connections by IP/port, `lpoptions` support and FFI bindings for the libcups3 DNS-SD API (PRs #18–#22, merged).
- libcups: fixed crashes in `cupsDNSSDDelete()` and a leaked client in the Avahi DNS-SD code (#164, #165, merged).

**Contributor — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy** | **Mar 2025 – Dec 2025**

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977, merged).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553, merged).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214, merged).

## TECHNICAL SKILLS

**Languages:** Rust, C, Modern C++, Python (asyncio), Bash, JavaScript (Node.js)

**Systems & Linux:** Linux (Ubuntu/Debian), operating systems, FFI and bindings, IPC (varlink, D-Bus; zbus, zlink), concurrency (async/event-driven, multithreading), memory management, networking (TCP/UDP), CUPS/IPP, DNS-SD/mDNS (Avahi), kernel modules and device drivers

**Tooling:** Cargo, Clippy, rustfmt, Git/GitHub, GitHub Actions CI, CMake, Make, Yocto, Buildroot, QEMU, testing, debugging, performance optimization, code review

**Desktop & embedded:** iced, libcosmic, PyQt5; Zephyr, FreeRTOS, ARM Cortex-M (STM32), ESP32

## ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024 (Egyptian Collegiate Programming Contest).
- Treasurer, IEEE Student Branch, Minia University: managed a budget of more than $10k.

## EDUCATION

**Bachelor of Science (B.Sc.), Computer and Systems Engineering** | **Minia University** | **2020 – 2025**

- Grade: Very Good with Honors. Graduation project: AUTOSAR-based V2X vehicle on STM32; built the firmware update path and a dual-bank bootloader with rollback.